S. No.	Concentration term	Expression	Illustration
7	Volume percentage (% v/v)	$\frac{\text{Volume of the solute (in mL)}}{\text{Volume of solution (in mL)}} \times 100$	50 mL of tincture of benzoin, an antiseptic solution contains 10 mL of benzoin. The volume percentage of benzoin $= \frac{\text{Volume of the benzoin (in mL)}}{\text{Volume of solution (in mL)}} \times 100$ $= \frac{10}{50} \times 100 = 20\%\ \text{v / v}$
8	Mass by volume percentage (% w/v)	$\frac{\text{Mass of the solute (in g)}}{\text{Volume of solution (in mL)}} \times 100$	A 60 mL of paracetamol paediatric oral suspension contains 3g of paracetamol. The mass percentage of paracetamol is $\frac{\text{Mass of the paracetamol (in g)}}{\text{Volume of solution (in mL)}} \times 100 =$ $= \frac{3}{60} \times 100 = 5\%\ \text{w / v}$
9	Parts per million (ppm)	$\frac{\text{Number of parts of the component}}{\text{Total number of parts of all components}} \times 10^6$ $= \frac{\text{Mass of the solute}}{\text{Mass of the solution}} \times 10^6$	50 g of tap water contains 20 mg of dissolved solids. The TDS value in ppm is $\frac{\text{Mass of the dissolved solids}}{\text{Mass of the water}} \times 10^6$ $\frac{20 \times 10^{-3}\ \text{g}}{50\ \text{g}} \times 10^6 = 400\ \text{ppm}$

Evaluate Yourself

1) If 5.6 g of KOH is present in (a) 500 mL and (b) 1 litre of solution, calculate the molarity of each of these solutions.
2) 2.82 g of glucose is dissolved in 30 g of water. Calculate the mole fraction of glucose and water.
3) The antiseptic solution of iodopovidone for the use of external application contains 10 % w/v of iodopovidone. Calculate the amount of iodopovidone present in a typical dose of 1.5 mL.
4) A litre of sea water weighing about 1.05 kg contains 5 mg of dissolved oxygen (O_2). Express the concentration of dissolved oxygen in ppm.

The concentration of a solution is expressed in different units. The choice of unit depends on the type of measurement applied. For example,in complexometric titrations involving

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